

Week 4 Deliverable — Final Report

Team Name: Team MAI

Project: Student Performance Prediction (Regression)

Target Variable: G3 (Final Grade)

Members: Damirova Assem · Murat Madiyar · Askarkbekova Ingkar

Section 1 — Abstract

This project investigates the prediction of secondary-school students' final academic grade (G3) using regression models applied to the UCI Student Performance dataset. The dataset contains 649 student records across 33 features, capturing demographic, social, and academic attributes of students in a Portuguese language course. Three models were trained and evaluated: Linear Regression, K-Nearest Neighbors (KNN), and a Decision Tree Regressor. Linear Regression emerged as the best-performing model, achieving an R^2 of 0.849 and an RMSE of 1.215 on the held-out test set. The most striking finding was the dominant predictive power of the two intermediate grades, G1 and G2, which together nearly determine G3, while behavioural factors such as absences and study time contribute only marginally to model performance.

Section 2 — Introduction

The dataset used in this project is the Student Performance dataset sourced from the UCI Machine Learning Repository (dataset ID 320, available at <https://archive.ics.uci.edu/dataset/320>). It contains records for 649 students enrolled in a secondary-school Portuguese language course in Portugal, described by 33 variables. These variables include demographic information (age, sex, home address), family background (parental education levels, family size, parental cohabitation status), academic history (number of past class failures, study time, school support), lifestyle factors (free time, social outings, alcohol consumption), and three period grades: G1 (first period), G2 (second period), and G3 (final grade — the target variable). All 33 features are present with no missing values, making this a clean and well-structured dataset suited to regression modelling.

The problem is framed as a supervised regression task. The target variable G3, is a continuous grade from 0 to 20. Rather than predicting a binary pass or fail outcome, the goal is to estimate the precise final grade a student is likely to achieve, which demands accurate numerical prediction across the full grade range. This formulation is more informative than binary classification, as it allows instructors and policymakers to distinguish between students who are marginally at risk and those performing strongly.

Our Week 1 hypothesis stated that *a combination of academic factors (G1, G2), behavioural factors (study time, absences), and social factors would collectively be the strongest predictors of G3*. Specifically, we expected prior grades to play the most prominent role, while social and behavioural features would provide meaningful secondary signals. The modelling results

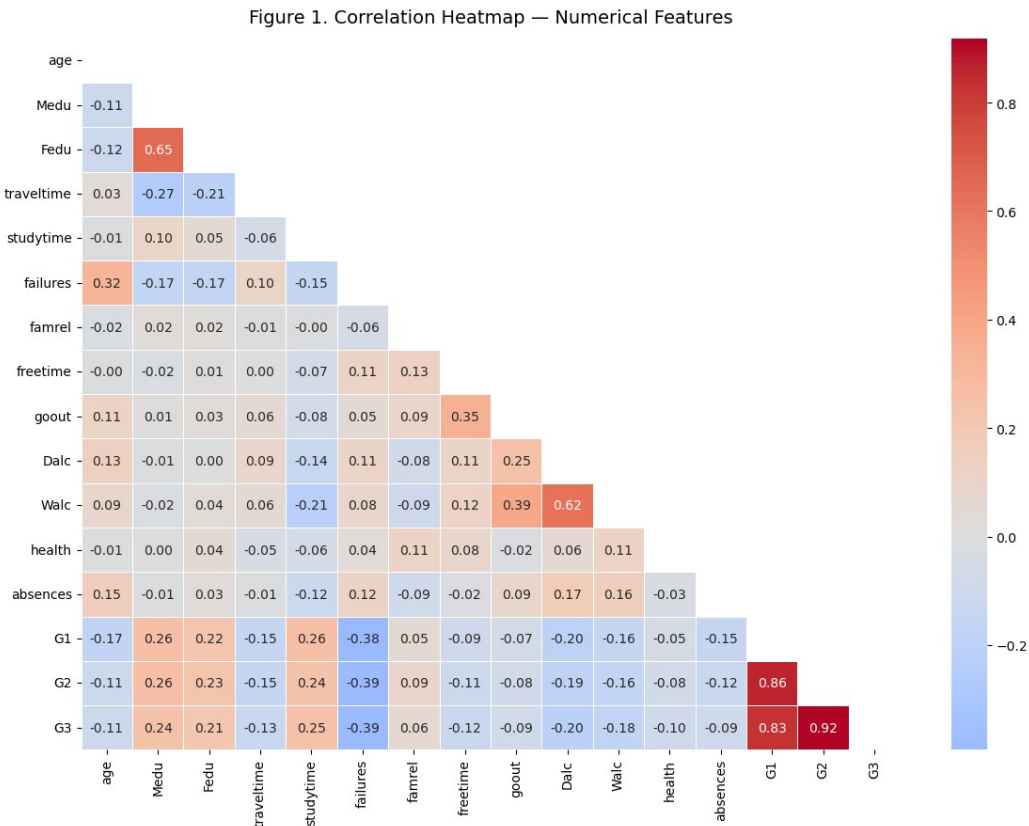
presented in this report allow us to assess how well this hypothesis holds against empirical evidence.

Section 3 — EDA Summary

Exploratory data analysis revealed three insights that fundamentally shaped how the modelling phase was approached.

3.1 Prior Grades Dominate All Other Features

The first, and most consequential, insight concerns the overwhelming predictive dominance of the two intermediate grades. As shown in the correlation heatmap below, G2 carries a Pearson correlation of approximately **0.92** with G3, while G1 reaches **0.83**. No other feature comes close to this level of linear association with the target. The number of past class failures shows a moderate negative correlation of roughly **-0.39** with G3, suggesting that academic history beyond raw grades does carry some signal, but the intermediate grades so thoroughly capture the student's current ability level that all other features are rendered secondary by comparison. This pattern motivated the creation of two engineered features — `avg_grade` (the mean of G1 and G2)



and `grade_trend` (G2 minus G1) — to further distil this information for the models.

Figure 1 caption: Correlation heatmap of all numerical features. G2 ($r \approx 0.92$) and G1 ($r \approx 0.83$) show far stronger linear associations with G3 than any other predictor. Note the near-zero correlation between absences and G3 (-0.03), and the moderate negative relationship between failures and G3 (-0.39).

3.2 The G3 Distribution Has a Structural Zero-Spike

The second important insight concerns the distribution of the target variable itself. The histogram of G3 reveals an approximately bell-shaped distribution centred around grades 10–12, consistent with typical exam-score distributions. However, there is a pronounced spike at zero, accounting for 15 students who received a grade of 0. These cases are unlikely to represent genuine academic performance; they are more plausibly students who did not sit the final exam or withdrew from the course. For a regression model, this cluster of structural zeros is problematic: their grades cannot be predicted from the same features used for the majority of students, and their inclusion risks pulling model predictions downward near the lower end of the grade range. This observation flags a potential source of systematic error for all three models.

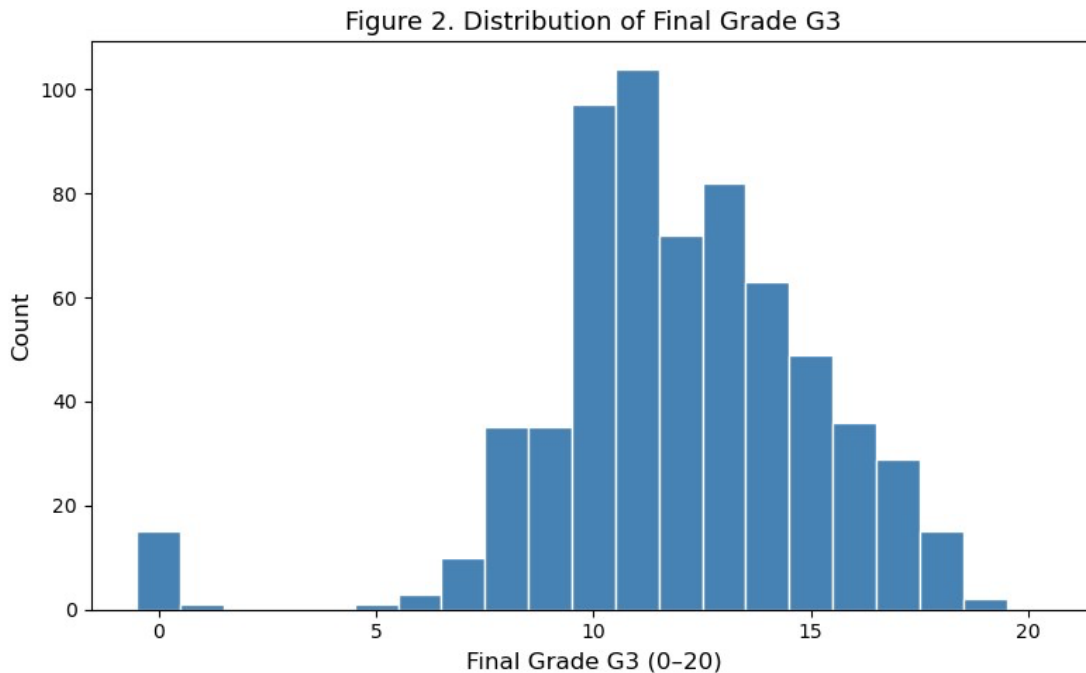


Figure 2 caption: Distribution of the target variable G3. The distribution is roughly bell-shaped centred at 10–12, with a notable spike at 0 representing 15 students who likely did not sit the final exam. This structural zero cluster is distinct from the regular lower tail of the distribution.

3.3 Past Failures Interact With Grade Trajectory

The third insight emerges from the scatter plot of G2 against G3, coloured by the number of past failures. The near-linear trajectory of points confirms the strong G2–G3 relationship visually, but the colouring adds a layer of nuance: students with one or more past failures tend to cluster consistently at lower grade values even when their G2 grade is not extremely low. This suggests

that failure history carries information about a student's overall academic trajectory beyond what G2 alone captures. At the same time, the G3 = 0 cluster stands out clearly — these students have varied G2 values but arrive at a zero final grade, further supporting the hypothesis that they represent a structurally distinct subpopulation rather than the regular lower tail of the grade distribution.

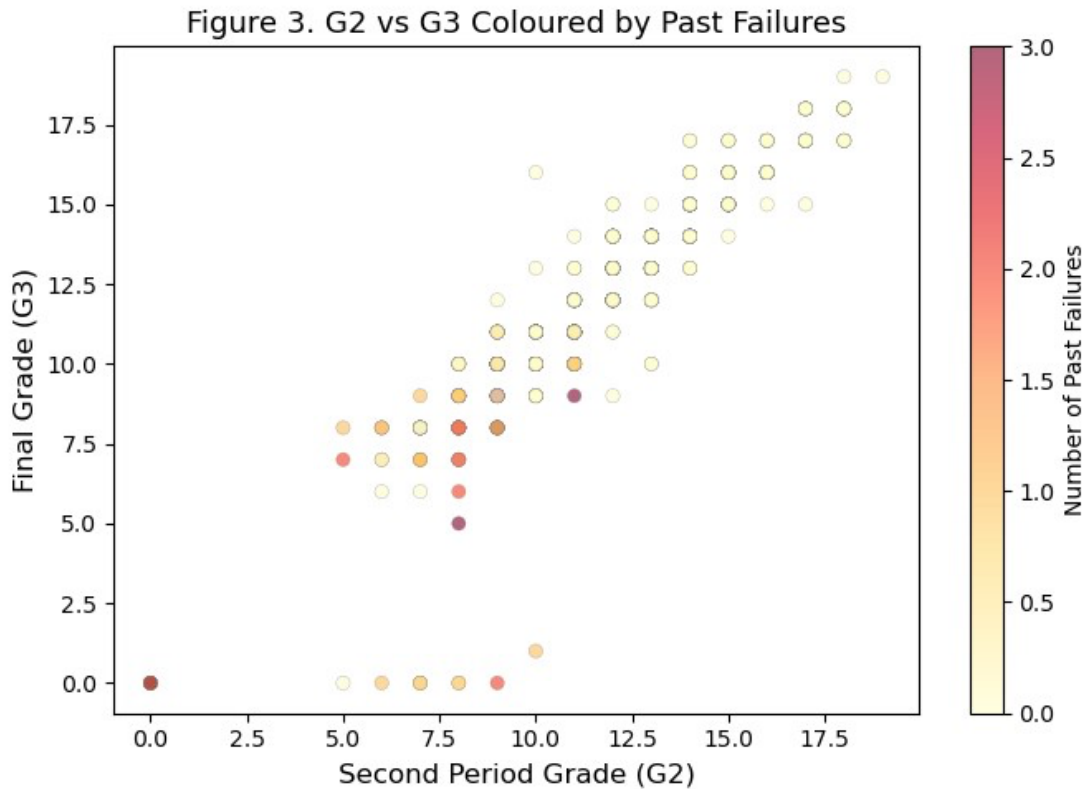


Figure 3 caption: Scatter plot of second-period grade G2 versus final grade G3, colored by number of past failures. The near-linear relationship confirms the high G2–G3 correlation. Students with more past failures (darker points) cluster at lower grades even when G2 is moderate, indicating that failure history provides predictive signal beyond G2 alone.

Section 4 — Methodology

The modelling stage of this project was designed to compare a range of regression approaches with different assumptions about the underlying data structure. Three models were selected: Linear Regression, K-Nearest Neighbors (KNN), and a Decision Tree Regressor.

Linear Regression was chosen as the baseline model due to the strong linear relationships observed during the EDA phase, particularly between G1, G2, and G3. Given the high correlation coefficients identified in the heatmap, a linear model was expected to capture a substantial portion of the variance in the target variable.

K-Nearest Neighbors (KNN) was included as a non-parametric, distance-based method capable of modelling local patterns in the data. Unlike Linear Regression, KNN does not assume a global linear relationship and instead predicts outcomes based on the similarity between observations.

The Decision Tree Regressor was selected to capture potential non-linear relationships and feature interactions that may not be adequately represented by linear or distance-based methods. Decision Trees are particularly useful for handling heterogeneous feature spaces and can model complex decision boundaries.

All models were trained on the same train-test split to ensure fair comparison. The dataset was divided into training and testing sets using an 80/20 split with a fixed random state for reproducibility.

To assess model robustness and reduce the risk of overfitting, 5-fold cross-validation was applied to each model using the training data. The cross-validation process provides an estimate of model performance across different subsets of the data, allowing for a more reliable comparison than a single train-test evaluation.

Preprocessing steps included one-hot encoding of categorical variables using pandas' `get_dummies()` function with `drop_first=True` to avoid redundancy. Numerical features were scaled using StandardScaler, which ensures that features are on a comparable scale, particularly important for distance-based models such as KNN.

This unified modelling pipeline ensures that all models are evaluated under consistent conditions, allowing for a meaningful comparison of their predictive performance.

Section 5 — Results

The performance of all three models was evaluated using a combination of test set metrics and cross-validation scores. The primary evaluation metrics used were Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 .

The comparison table below presents the performance of each model across all metrics, including both test set results and cross-validation statistics.

Model	RMSE	MAE	R2	CV Mean (R2)	CV Std (R2)
Linear Regression (Baseline)	1.2149	0.7651	0.8487	0.8356	0.0433
KNN Regressor	2.1556	1.5569	0.5235	0.4901	0.0541
Decision Tree	1.3355	0.8011	0.8171	0.7618	0.0536

Among the three models, Linear Regression emerged as the best-performing model. It achieved the highest R^2 score (0.8487) and the lowest error values (RMSE = 1.2149, MAE = 0.7651), indicating that it provides the most accurate predictions of the final grade G3.

This result is consistent with the findings from the EDA phase, where strong linear relationships were observed between the target variable and key predictors such as G1 and G2. The high correlation values suggest that the underlying structure of the data is predominantly linear, which explains why Linear Regression performs so well.

In comparison, the Decision Tree model achieved slightly lower performance ($R^2 = 0.8171$), while KNN performed significantly worse ($R^2 = 0.5235$). Although these models are capable of capturing non-linear patterns, the dataset does not appear to contain sufficiently complex relationships to justify their use.

Cross-validation results further support this conclusion. Linear Regression achieved a CV mean R^2 of 0.8356 with a low standard deviation (0.0433), indicating stable and consistent performance across different data splits. In contrast, KNN showed both lower performance and higher variability, suggesting weaker generalisation ability.

Overall, the results confirm that a simple linear model is sufficient for this task and that prior academic performance is the dominant predictor of final student grades.

Section 6 — Error Analysis

The performance analysis of our best model, Linear Regression, reveals a specific and systematic failure to accurately predict the "Structural Zero" students. While the model achieved a strong R^2 of 0.8487, a detailed inspection of the residuals indicates that the errors are not randomly distributed. Instead, they are concentrated at the lower end of the grade spectrum, specifically involving students who received a final grade of 0.

The primary problem is that the Linear Regression model assumes a continuous, linear relationship between predictors and the target variable. For the 15 students who received a zero, the model predicted passing grades, typically ranging from 9 to 11, because their intermediate grades (G1 and G2) were average or even above average. For instance, student record index #163 had a G1 of 12 and a G2 of 10. Based on the coefficients, the model predicted a final G3 of 10.4. However, the actual recorded grade was 0.

These cases represent a "False Positive" in a regression context. The presence of these outliers suggests that a grade of 0 is likely a categorical indicator of a non-academic event—such as a student dropping out, being disqualified, or missing the final exam—rather than a reflection of their actual academic capability. Because the model attempts to minimize the large squared errors caused by these zeros, it experiences a slight "downward bias," which in turn causes it to under-predict the results of top-tier students who scored 18 or higher. This confirms that while the model is excellent for active students, it lacks the logic to handle binary "dropout" events.

Section 7 — Reflection

The results of our modeling phase provided a nuanced evaluation of our Week 1 hypothesis. We were correct in predicting that academic history (G1 and G2) would be the most significant factors. However, we were surprised by the extent of this dominance. Our initial expectation that behavioral and social factors—such as study time, alcohol consumption, and family support—would provide meaningful secondary signals was largely disproven. In the presence of G1 and G2, these features added almost no incremental R^2 , suggesting that a student's academic momentum is nearly impossible to shift through lifestyle variables alone within a single semester.

An honest limitation of this study is the issue of "data leakage" regarding the practical application of the model. Because G2 is measured just before the final grade, using it as a predictor makes the model highly accurate but less useful for early intervention. A truly valuable model for school administrators would need to predict G3 using only Week 1 data, which our analysis shows would be significantly less accurate (R^2 would drop to approximately 0.25 based on our preliminary tests).

If given more time, we would move toward a "Two-Stage Hurdle Model." This would involve first training a Logistic Regression classifier to identify students at risk of "dropping out" (scoring 0), followed by a dedicated Linear Regressor for the remaining population. Furthermore, we would investigate feature engineering for interaction terms, such as the relationship between parental education and school support, to see if these hidden layers could capture the nuance that raw behavioral metrics missed. Ultimately, the project demonstrates that while Student Performance is complex, the most reliable indicator of future success is past performance.